

PRE-CLASS READING 5 OF 5

A Path Towards Autonomous Machine Intelligence

Yann LeCun (2022). *OpenReview position paper, version 0.9.2, 27 June 2022.*

ASSIGNED READING

Abstract + Sections 1–3 — the six-module architecture and JEPa

~18 of 62 pp.

Technical appendices on training procedures — optional

skip

≈ 40 minutes

Note on what this is. This is a position paper, not a peer-reviewed result — LeCun's own proposal for a research direction, not something built and proven. It's also nearly four years old. Some of what it argues against (that language models can't plan or reason) looks different today than it did in 2022, given what “reasoning models” can now do. Read it as one credible researcher's bet on a different path, not as settled fact — and hold it up against what actually happened.

Week 1 pre-reading is heavier than most. Across all five readings this week, budget **4–5 hours**, not the usual 3. Later weeks are lighter to compensate. This is the foundational week — worth the extra time.

KEY TAKEAWAY

LeCun argues that scaling up today's dominant approach — training a model to predict the next word or pixel — will not by itself produce a machine that can reason, plan, or learn as efficiently as a human toddler. His proposed fix isn't a bigger model; it's a different architecture built around a “world model” that predicts consequences of actions before taking them.

WHY IT MATTERS FOR THIS COURSE

This reading is deliberately the odd one out. Readings 1, 3 and 4 all sit inside the generative/foundation-model lineage that dominates this field right now. LeCun is arguing that lineage is insufficient — a genuinely different bet from one of the field's most credited researchers (he shares the 2018 Turing Award for deep learning with Geoffrey Hinton and Yoshua Bengio). Holding a real disagreement between experts in your head, rather than assuming the field has one settled answer, is exactly the habit Class 1's taxonomy is trying to build.

MAIN POINTS

The problem LeCun is trying to solve

- **The efficiency gap.** A teenager learns to drive competently in about 20 hours of practice. Training a self-driving system to a comparable standard takes vastly more data than any human ever sees. LeCun's claim: this gap exists because humans and animals build an internal model of how the world works — physics, cause and effect, what happens if I do this — and machines mostly don't.
- **Three questions the paper is organized around:** How could machines learn as efficiently as humans and animals? How could machines learn to reason and plan? How could machines learn representations of the world at multiple levels of abstraction, so they can plan over both short and long time horizons?

The six-module architecture

LeCun proposes an agent built from six interacting, trainable modules:

- **Configurator** — sets up the other modules for the task at hand, the way a manager might brief a team before a specific job.
- **Perception** — takes in raw sensory input and estimates the current state of the world.
- **World model** — the centerpiece. Predicts what will happen next, and what would happen under different possible actions, without actually taking them. This is what lets an agent imagine consequences before acting.
- **Cost** — a built-in sense of “how good is this situation,” split into a fixed intrinsic part (hard-wired discomfort/reward signals) and a trainable part that learns to anticipate cost further into the future.
- **Actor** — proposes a candidate action, or sequence of actions, and picks the one the world model + cost module predict will turn out best.

- **Short-term memory** — tracks the current situation and its evaluated cost, giving the system continuity across a sequence of decisions.
- **How they fit together, step by step:** perceive the current state → propose several candidate actions → use the world model to simulate what each one leads to → use the cost module to score those imagined outcomes → the actor picks the lowest-cost action → act → update short-term memory → repeat.

JEPA — how the world model actually learns

- **The core idea.** A Joint Embedding Predictive Architecture doesn't try to predict raw pixels or exact future frames. It predicts in a compressed, abstract representation — predicting *what matters* about the future, not *every detail* of it.
“The details of the texture on a carpet, the leaves of a tree moving in the wind, or the ripples on a pond, cannot be predicted accurately, at least not over long time periods and not without consuming enormous resources.”
- A model forced to predict every pixel wastes capacity on unpredictable noise. JEPA is built to ignore what can't be predicted and focus on what can.
- **Stacking for different timescales (H-JEPA).** The paper's own example: a cook making crêpes. At a high level, you can predict “fetch flour, milk, and eggs” — a plan good for tens of seconds. At a low level, you'd need to predict the exact millisecond-by-millisecond trajectory of the cook's hands — only predictable a fraction of a second ahead. Stacking JEPAs at different levels of abstraction lets one system make confident long-range plans at the top and precise short-range predictions at the bottom, simultaneously.

Where this disagrees with the rest of your reading list

- **On language models (bearing directly on Reading 3).** LeCun's position is that predicting the next token — exactly what Karpathy's talk describes as the entire mechanism of an LLM — is fundamentally the wrong kind of prediction for building real world understanding, because it predicts in the wrong space (raw tokens, not abstract structure) and has no built-in mechanism for planning ahead.
- **On reinforcement learning.** LeCun argues reward-driven trial-and-error learning, the dominant paradigm in most robotics and game-playing AI at the time, is far too sample-inefficient to explain how humans and animals actually learn. Most of what a child learns, in his view, comes from unsupervised prediction of what happens next — not from being rewarded or punished.
- **A useful contrast with Reading 4.** AlphaFold is arguably a working, narrow example of exactly the kind of “world model” LeCun wants generalized — a system that predicts a structural outcome (a folded protein) accurately enough to act on. The open question this paper poses is whether that kind of success can generalize beyond a single, well-defined scientific problem into something that plans and reasons broadly.

READING FOCUS QUESTIONS — bring your answers to class

There are no wrong answers here.

1. LeCun says a teenager learns to drive well in about 20 hours, while machines need vastly more experience to do the same job. Think of something *you* learned quickly from very little practice, and something that took you an enormous number of repetitions to get right. What made the difference?
2. The crêpe example splits planning into a high level (“fetch the ingredients”) and a low level (the exact hand movements). Pick something you know how to do well — cooking a dish, playing an instrument, a sport — and describe it at both levels. Which level would be easier to teach to someone else, and why?
3. This paper is nearly four years old, and it made a specific bet: that predicting the next word (what today's LLMs do) is the wrong foundation for machines that can truly reason and plan. Based on everything else you've read this week, does that bet look right, wrong, or still open? What evidence would change your mind either way?

SOURCE LINKS

Paper · [OpenReview — A Path Towards Autonomous Machine Intelligence](#) · [Direct PDF](#)

LeCun's own talk on the same material · [YouTube](#)

CAVEATS

- 1. This summary is not a substitute for the reading.** It is a map, not the territory. This is a discursive, essay-style paper — LeCun's own diagrams and worked examples land better read directly than summarized.
- 2. This summary was generated and fact-checked using Anthropic models,** cross-referenced against three independent technical reviews of the paper and the OpenReview listing itself (title, date, version, abstract confirmed directly). This is a position paper expressing one researcher's view, not a peer-reviewed finding — treat its claims as arguable, not settled. Errors are still possible. Flag anything that looks wrong.